JESS Thermodynamic Database v8.9

Chemical species in reactions with Promethium

24-Jan-23

Charge	CAS	Count	Molecular formula	Mol. mass
[Pent]11OHCOO-1				
-1	---	69	C(6)H(9)O(3)	129.136
2OH2MePropan-1				
-1	---	197	C(4)H(7)O(3)	103.098
CDTA-4 trans-1,2-cyclohexylenedinitrilotetraacetate ion; trans-1,2-diaminocyclohexanetetraacetate ion; CDTA ion				
-4	22005-54-5	301	C(14)H(18)N(2)O(8)	342.306
Citric-3 Citrate ion				
-3	126-44-3	1347	C(6)H(5)O(7)	189.102
Cl-1 Chloride ion				
-1	16887-00-6	2242	Cl(1)	35.4530
CO3-2 Carbonate ion				
-2	3812-32-6	1435	C(1)O(3)	60.0092
DETAP-4				
-4	---	36	C(13)H(18)N(2)O(9)	346.294
e-1 Electron				
-1	---	3399	E(1)	0.000000
EDTA-4 Ethylenediaminetetraacetate anion				
-4	---	1457	C(10)H(12)N(2)O(8)	288.214

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F-1
Fluoride ion

-1	16984-48-8	1090	F(1)	18.9984

Gly-1
Glycinate ion; Aminoacetate ion

-1	---	1356	C(2)H(4)N(1)O(2)	74.0593

H+1
Hydrogen ion; Proton

1	12408-02-5	31679	H(1)	1.00794

H+1_Citric-3

-2	---	93	C(6)H(6)O(7)	190.109

H+1_CO3-2

-1	---	253	C(1)H(1)O(3)	61.0171

H+1_DETAP-4

-3	---	18	C(13)H(19)N(2)O(9)	347.302

H+1_Gly-1
Glycine; Aminoacetic acid; Aminoethanoic acid; 2-Aminoethanoic acid

0	---	82	C(2)H(5)N(1)O(2)	75.0672

H+1_MePhos-2

-1	---	6	C(1)H(4)O(3)P(1)	95.0150

H+1_OHMePhos-2

-1	---	8	C(1)H(4)O(4)P(1)	111.014

H+1_PhosAcet-3

-2	---	21	C(2)H(3)O(5)P(1)	138.017

H+1_PO4-3
Hydrogenphosphate ion

-2	---	317	H(1)O(4)P(1)	95.9795

H+1(2)_EDDPI-2
Ethylenebis(iminomethylenephosphinic acid); EDDPI; Ethylenediamine-N,N'-di(methylenephosphinic) acid

0	---	5	C(4)H(14)N(2)O(4)P(2)	216.114

H+1(2)_Etbis(ImMePhos)-4

-2	---	22	C(4)H(12)N(2)O(6)P(2)	246.097

H+1(2)_PhosAcet-3

-1	---	5	C(2)H(4)O(5)P(1)	139.025

H+1(2)_PO4-3
Dihydrogenphospate ion

-1	14066-20-7	284	H(2)O(4)P(1)	96.9875

H2O
Water

0	7732-18-5	5947	H(2)O(1)	18.0153

HIMDA-2

-2	---	201	C(6)H(9)N(1)O(5)	175.141

IO3-1
Iodate ion

-1	15454-31-6	291	I(1)O(3)	174.898

Lactic-1
Lactate ion

-1	---	643	C(3)H(5)O(3)	89.0709

Mandelic-1
alpha-Hydroxybenzeneactate ion; Mandelate ion; L-Phenylhydroxyacetate ion; Hydroxybenzene acetate ion; Phenylglycolate ion; Uromalinate ion; Amaygdalate ion

-1	769-61-9	132	C(8)H(7)O(3)	151.142

N3-1
Azide ion

-1	14343-69-2	176	N(3)	42.0201

NO3-1
Nitrate ion

-1	14797-55-8	573	N(1)O(3)	62.0049

NTA-3
Nitrilotriacetate ion; N,N-bis(carboxymethyl)glycinate ion; Triglycollamate ion; Triglycinate ion; NTA ion

-3	28528-44-1	751	C(6)H(6)N(1)O(6)	188.117

O2
Oxygen (aquated)

0	---	67	O(2)	31.9988

O2(g)
Oxygen gas; Oxygen

0	7782-44-7	2559	O(2)	31.9988

OH-1
Hydroxide ion

-1	14280-30-9	6558	H(1)O(1)	17.0073

Oxalic-2
Oxalate ion

-2	338-70-5	1241	C(2)O(4)	88.0196

Oxine-1
8-Hydroxyquinolate ion; Oxine ion

-1	16582-16-4	162	C(9)H(6)N(1)O(1)	144.153

P3O9-3
Trimetaphosphate ion

-3	---	22	O(9)P(3)	236.917

Pm+2
Promethium(II) ion

2	---	3	Pm(1)	145.000

Pm+3
Promethium(III) ion

3	22541-16-8	84	Pm(1)	145.000

Pm+3_[Pent]11OHCOO-1

2	---	2	C(6)H(9)O(3)Pm(1)	274.136

Pm+3_[Pent]11OHCOO-1(2)

1	---	2	C(12)H(18)O(6)Pm(1)	403.271

Pm+3_[Pent]11OHCOO-1(3)

0	---	1	C(18)H(27)O(9)Pm(1)	532.407

Pm+3_2OH2MePropan-1

2	---	1	C(4)H(7)O(3)Pm(1)	248.098

Pm+3_2OH2MePropan-1(2)

1	---	1	C(8)H(14)O(6)Pm(1)	351.196

Pm+3_2OH2MePropan-1(3)

0	---	1	C(12)H(21)O(9)Pm(1)	454.293

Pm+3_CDTA-4

-1	---	1	C(14)H(18)N(2)O(8)Pm(1)	487.306

Pm+3_Citric-3

0	---	3	C(6)H(5)O(7)Pm(1)	334.102

Pm+3_Citric-3(2)

-3	---	2	C(12)H(10)O(14)Pm(1)	523.203

Pm+3_Cl-1

2	---	2	Cl(1)Pm(1)	180.453

Pm+3_Cl-1(2)

1	---	1	Cl(2)Pm(1)	215.906

Pm+3_CO3-2

1	---	1	C(1)O(3)Pm(1)	205.009

Pm+3_CO3-2(2)

-1	---	1	C(2)O(6)Pm(1)	265.018

Pm+3_DETAP-4

-1	---	1	C(13)H(18)N(2)O(9)Pm(1)	491.294

Pm+3_EDTA-4

-1	---	1	C(10)H(12)N(2)O(8)Pm(1)	433.214

Pm+3_F-1

2	---	1	F(1)Pm(1)	163.998

Pm+3_F-1(2)

1	---	1	F(2)Pm(1)	182.997

Pm+3_H+1_Citric-3

1	---	3	C(6)H(6)O(7)Pm(1)	335.109

Pm+3_H+1_Citric-3(2)

-2	---	3	C(12)H(11)O(14)Pm(1)	524.211

Pm+3_H+1_CO3-2

2	---	1	C(1)H(1)O(3)Pm(1)	206.017

Pm+3_H+1_DETAP-4

0	---	1	C(13)H(19)N(2)O(9)Pm(1)	492.302

Pm+3_H+1_Gly-1

3	---	2	C(2)H(5)N(1)O(2)Pm(1)	220.067

Pm+3_H+1_MePhos-2

2	---	1	C(1)H(4)O(3)P(1)Pm(1)	240.015

Pm+3_H+1_OHMePhos-2

2	---	1	C(1)H(4)O(4)P(1)Pm(1)	256.014

Pm+3_H+1_PhosAcet-3				
1	---	1	C(2)H(3)O(5)P(1)Pm(1)	283.017
Pm+3_H+1_PO4-3				
1	---	2	H(1)O(4)P(1)Pm(1)	240.980
Pm+3_H+1(2)_Citric-3(2)				
-1	---	2	C(12)H(12)O(14)Pm(1)	525.219
Pm+3_H+1(2)_EDDPI-2				
3	---	1	C(4)H(14)N(2)O(4)P(2)Pm(1)	361.114
Pm+3_H+1(2)_Etbis(ImMePhos)-4				
1	---	1	C(4)H(12)N(2)O(6)P(2)Pm(1)	391.097
Pm+3_H+1(2)_OHMePhos-2(2)				
1	---	1	C(2)H(8)O(8)P(2)Pm(1)	367.029
Pm+3_H+1(2)_PhosAcet-3				
2	---	1	C(2)H(4)O(5)P(1)Pm(1)	284.025
Pm+3_H+1(2)_PhosAcet-3(2)				
-1	---	1	C(4)H(6)O(10)P(2)Pm(1)	421.034
Pm+3_H+1(2)_PO4-3				
2	---	2	H(2)O(4)P(1)Pm(1)	241.988
Pm+3_H+1(2)_PO4-3(2)				
-1	---	2	H(2)O(8)P(2)Pm(1)	336.959
Pm+3_HIMDA-2				
1	---	1	C(6)H(9)N(1)O(5)Pm(1)	320.141
Pm+3_IO3-1				
2	---	1	I(1)O(3)Pm(1)	319.898

Pm+3_IO3-1(3)_(s)

0	---	1	I(3)O(9)Pm(1)	669.695

Pm+3_Lactic-1

2	---	2	C(3)H(5)O(3)Pm(1)	234.071

Pm+3_Lactic-1(2)

1	---	2	C(6)H(10)O(6)Pm(1)	323.142

Pm+3_Lactic-1(3)

0	---	1	C(9)H(15)O(9)Pm(1)	412.213

Pm+3_Mandelic-1

2	---	1	C(8)H(7)O(3)Pm(1)	296.142

Pm+3_Mandelic-1(2)

1	---	1	C(16)H(14)O(6)Pm(1)	447.284

Pm+3_Mandelic-1(3)

0	---	1	C(24)H(21)O(9)Pm(1)	598.425

Pm+3_N3-1

2	---	1	N(3)Pm(1)	187.020

Pm+3_NO3-1

2	---	1	N(1)O(3)Pm(1)	207.005

Pm+3_NTA-3

0	---	2	C(6)H(6)N(1)O(6)Pm(1)	333.117

Pm+3_NTA-3(2)

-3	---	2	C(12)H(12)N(2)O(12)Pm(1)	521.234

Pm+3_OH-1

2	---	3	H(1)O(1)Pm(1)	162.007

Pm+3_OH-1(2)

1	---	1	H(2)O(2)Pm(1)	179.015

Pm+3_OH-1(3)

0	---	1	H(3)O(3)Pm(1)	196.022

Pm+3_OH-1(3)_(s)

Promethium hydroxide

0	---	3	H(3)O(3)Pm(1)	196.022

Pm+3_Oxalic-2

1	---	2	C(2)O(4)Pm(1)	233.020

Pm+3_Oxalic-2(2)

-1	---	2	C(4)O(8)Pm(1)	321.039

Pm+3_Oxalic-2(3)

-3	---	1	C(6)O(12)Pm(1)	409.059

Pm+3_Oxine-1

2	---	1	C(9)H(6)N(1)O(1)Pm(1)	289.153

Pm+3_Oxine-1(2)

1	---	1	C(18)H(12)N(2)O(2)Pm(1)	433.306

Pm+3_Oxine-1(3)

0	---	1	C(27)H(18)N(3)O(3)Pm(1)	577.458

Pm+3_P3O9-3

0	---	1	O(9)P(3)Pm(1)	381.917

Pm+3_PO4-3

0	---	1	O(4)P(1)Pm(1)	239.972

Pm+3_PO4-3(2)

-3	---	1	O(8)P(2)Pm(1)	334.943

Pm+3_SO4-2

1	---	1	O(4)Pm(1)S(1)	241.058

Pm+3_SO4-2(2)

-1	---	1	O(8)Pm(1)S(2)	337.115

Pm+3_Tartaric-2

1	---	1	C(4)H(4)O(6)Pm(1)	293.072

Pm+3_Tartaric-2(2)

-1	---	1	C(8)H(8)O(12)Pm(1)	441.144

Pm(s)

Promethium

0	---	5	Pm(1)	145.000

Pm2O3(s)

Promethium Oxide

0	---	2	O(3)Pm(2)	337.998

PO4-3

Phosphate ion; Phosfate ion

-3	14265-44-2	1594	O(4)P(1)	94.9716

SO4-2

Sulfate ion; Sulphate ion

-2	14808-79-8	1270	O(4)S(1)	96.0576

Tartaric-2

Tartrate ion

-2	---	393	C(4)H(4)O(6)	148.072